Technical Design Document: KajaAiken_ProgrammingExercise_1

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Program Description: This program automates a ticket pre-sale system by managing a strict total of 20 tickets. It loops continuously to prompt users for purchase amounts, validating that inputs are valid whole numbers, don't exceed a cap of 4 tickets, and don't exceed the remaining stock. Once sold out, it outputs the total number of unique buyers.

Function: main_program_logic

Executes the ticket sales loop, processes user requests, handles bad inputs, and tracks overall metrics.

Parameters:

None

Variables:

TOTAL_TICKETS (int): Constant setting the initial pre-sale inventory to 20.

MAX_TICKETS_PER_BUYER (int): Constant enforcing the maximum 4 ticket limit per person.

remaining_tickets (int): Accumulator tracking how many tickets are left for sale.

total_buyers (int): Tracker counting the total number of successful buyers.

requested_tickets (int): Integer capturing the user's desired ticket purchase amount.

Logic:

Initialize TOTAL_TICKETS to 20 and MAX_TICKETS_PER_BUYER to 4.

Set remaining_tickets equal to TOTAL_TICKETS.

Set total_buyers to 0.

Start a while loop that continues as long as remaining_tickets is greater than 0.

Display the number of remaining_tickets available.

Open a try-except block to catch input formatting errors.

Prompt the user to enter how many tickets they want to buy and convert it to an integer (requested_tickets).

If requested_tickets is greater than MAX_TICKETS_PER_BUYER, display an error message.

Elif if requested_tickets is greater than remaining_tickets, display an inventory limit error message.

Elif if requested_tickets is less than or equal to 0, display an invalid amount error message.

Else (the purchase is valid):

Subtract requested_tickets from remaining_tickets.

Increment total_buyers by 1.

Display a thank you message and show updated remaining tickets.

If a ValueError occurs during input conversion, display a message asking for a valid whole number.

Once the loop ends (remaining_tickets hits 0), display the final pre-sale closing announcement and the total count of total_buyers.

Return:

None

Link to your repository: https://github.com/kajaoe

Output Screenshot: